

NYCU Introduction to Machine Learning, Homework 1

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Part. 1, Coding (60%):

(10%) Linear Regression Model - Closed-form Solution

1. (10%) Show the weights and intercepts of your linear model.

All upper images are “No standardization, mini-batch GD” :A.

All lower image are “Standardization, batch GD” : B.

A.

```
2025-09-30 21:01:20.221 | INFO | __main__:main:179 - LR_CF.weights=array([2.85501274, 1.01785863, 0.47202168, 0.19608925]),  
LR_CF.intercept=-33.6956
```

B.

```
2025-09-30 21:56:49.174 | INFO | __main__:main:160 - LR_CF.weights=array([ 7.3935217 , 17.59620753, 0.80104512, 0.5638124  
1]), LR_CF.intercept=55.2859
```

(40%) Linear Regression Model - Gradient Descent Solution

2. (10%)

- Show the hyperparameters of your setting (e.g., learning rate, number of epochs, batch size, etc.).
- Show the weights and intercepts of your linear model.

A.

Learning_rate is not 0.01, the real lr is following the formula below.

```
LR_GD.fit(train_x, train_y, learning_rate=0.01, epochs=20000)  
batch_size = 32
```

Steepest Descent with Exact Line Search optimal step size

```
r = (mn_b_y - b_val)  
  
lr = (r.T @ r) / (r.T @ mn_b_X @ mn_b_X.T @ r) / batch_size  
  
2025-09-30 21:25:45.584 | INFO | __main__:main:188 - LR_GD.weights=array([2.84955367, 1.01744672, 0.46927591, 0.19935565]),  
LR_GD.intercept=-33.6794
```

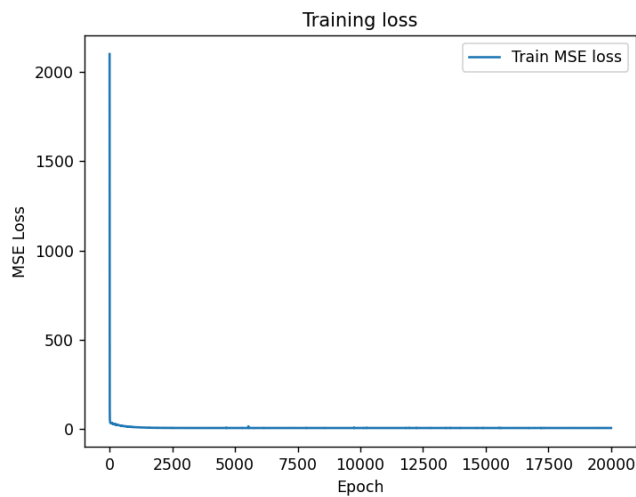
B.

Learning_rate is 0.01.

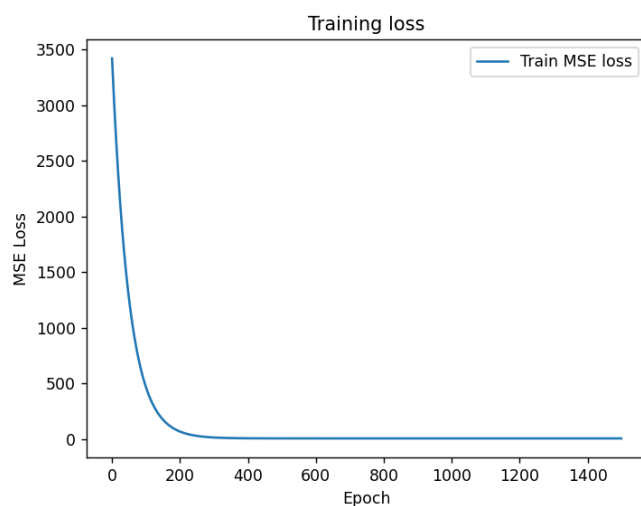
```
LR_GD.fit(train_x, train_y, learning_rate=0.01, epochs=1500)  
  
2025-09-30 21:56:49.211 | INFO | __main__:main:169 - LR_GD.weights=array([ 7.39351786, 17.59620156, 0.8010456 , 0.5638139  
1]), LR_GD.intercept=55.2859
```

3. (10%) Plot the learning curve. (x-axis=epoch, y-axis=training loss)

A.



B.



4. (20%) Show your MSE.cf, MSE.gd, and error rate between your closed-form solution and the gradient descent solution.

A.

```
2025-09-30 21:07:29.345 | INFO | __main__:main:206 - Prediction difference: 0.0430
2025-09-30 21:07:29.345 | INFO | main :main:211 - mse_cf=4.2903, mse_gd=4.2927. Difference: 0.056%
```

B.

```
2025-09-30 21:58:35.793 | INFO | __main__:main:187 - Prediction difference: 0.0000
2025-09-30 21:58:35.793 | INFO | main :main:192 - mse_cf=4.2903, mse_gd=4.2903. Difference: 0.000%
```

(10%) Code Check and Verification

5. (10%) Lint the code and show the PyTest results.

A.

```
PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\hw1> uv run flake8 .\main.py
PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\hw1>
```

```

PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\hw1> uv run pytest .\test_main.py -s
===== test session starts =====
platform win32 -- Python 3.12.2, pytest-8.4.2, pluggy-1.6.0
rootdir: C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\hw1
configfile: pyproject.toml
collected 2 items

test_main.py (100, 1)
2025-10-01 13:29:52.436 | INFO | test_main:test_regression_cf:30 - model.weights=array([[3.]]) , model.intercept=array([4.])
.(100, 1)
2025-10-01 13:29:52.436 | INFO | main:fit:125 - EPOCH 0, loss=30443.0888, lr=0.0000
2025-10-01 13:29:52.475 | INFO | main:fit:125 - EPOCH 1000, loss=86.5339, lr=0.0000
2025-10-01 13:29:52.516 | INFO | main:fit:125 - EPOCH 2000, loss=9.2983, lr=0.0000
2025-10-01 13:29:52.555 | INFO | main:fit:125 - EPOCH 3000, loss=4.4054, lr=0.0000
2025-10-01 13:29:52.595 | INFO | main:fit:125 - EPOCH 4000, loss=2.2080, lr=0.0000
2025-10-01 13:29:52.632 | INFO | main:fit:125 - EPOCH 5000, loss=1.0779, lr=0.0007
2025-10-01 13:29:52.668 | INFO | main:fit:125 - EPOCH 6000, loss=0.5492, lr=0.0000
2025-10-01 13:29:52.702 | INFO | main:fit:125 - EPOCH 7000, loss=0.2638, lr=0.0009
2025-10-01 13:29:52.736 | INFO | main:fit:125 - EPOCH 8000, loss=0.1260, lr=0.0000
2025-10-01 13:29:52.783 | INFO | main:fit:125 - EPOCH 9000, loss=0.0610, lr=0.0000
2025-10-01 13:29:52.818 | INFO | main:fit:125 - EPOCH 10000, loss=0.0283, lr=0.0001
2025-10-01 13:29:52.853 | INFO | main:fit:125 - EPOCH 11000, loss=0.0134, lr=0.0000
2025-10-01 13:29:52.885 | INFO | main:fit:125 - EPOCH 12000, loss=0.0060, lr=0.0001
2025-10-01 13:29:52.925 | INFO | main:fit:125 - EPOCH 13000, loss=0.0028, lr=0.0000
2025-10-01 13:29:52.956 | INFO | main:fit:125 - EPOCH 14000, loss=0.0014, lr=0.0000
2025-10-01 13:29:52.998 | INFO | main:fit:125 - EPOCH 15000, loss=0.0007, lr=0.0000
2025-10-01 13:29:53.044 | INFO | main:fit:125 - EPOCH 16000, loss=0.0003, lr=0.0000
2025-10-01 13:29:53.085 | INFO | main:fit:125 - EPOCH 17000, loss=0.0002, lr=0.0000
2025-10-01 13:29:53.119 | INFO | main:fit:125 - EPOCH 18000, loss=0.0001, lr=0.0000
2025-10-01 13:29:53.155 | INFO | main:fit:125 - EPOCH 19000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.185 | INFO | main:fit:125 - EPOCH 20000, loss=0.0000, lr=0.0004
2025-10-01 13:29:53.234 | INFO | main:fit:125 - EPOCH 21000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.267 | INFO | main:fit:125 - EPOCH 22000, loss=0.0000, lr=0.0010
2025-10-01 13:29:53.305 | INFO | main:fit:125 - EPOCH 23000, loss=0.0000, lr=0.0002
2025-10-01 13:29:53.344 | INFO | main:fit:125 - EPOCH 24000, loss=0.0000, lr=0.0002
2025-10-01 13:29:53.381 | INFO | main:fit:125 - EPOCH 25000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.416 | INFO | main:fit:125 - EPOCH 26000, loss=0.0000, lr=0.0001
2025-10-01 13:29:53.450 | INFO | main:fit:125 - EPOCH 27000, loss=0.0000, lr=0.0002
2025-10-01 13:29:53.495 | INFO | main:fit:125 - EPOCH 28000, loss=0.0000, lr=0.0001
2025-10-01 13:29:53.533 | INFO | main:fit:125 - EPOCH 29000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.567 | INFO | main:fit:125 - EPOCH 30000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.605 | INFO | main:fit:125 - EPOCH 31000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.643 | INFO | main:fit:125 - EPOCH 32000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.680 | INFO | main:fit:125 - EPOCH 33000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.716 | INFO | main:fit:125 - EPOCH 34000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.755 | INFO | main:fit:125 - EPOCH 35000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.790 | INFO | main:fit:125 - EPOCH 36000, loss=0.0000, lr=0.0002
2025-10-01 13:29:53.825 | INFO | main:fit:125 - EPOCH 37000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.862 | INFO | main:fit:125 - EPOCH 38000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.898 | INFO | main:fit:125 - EPOCH 39000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.933 | INFO | main:fit:125 - EPOCH 40000, loss=0.0000, lr=0.0000
2025-10-01 13:29:53.968 | INFO | main:fit:125 - EPOCH 41000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.005 | INFO | main:fit:125 - EPOCH 42000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.036 | INFO | main:fit:125 - EPOCH 43000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.078 | INFO | main:fit:125 - EPOCH 44000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.107 | INFO | main:fit:125 - EPOCH 45000, loss=0.0000, lr=0.0001
2025-10-01 13:29:54.145 | INFO | main:fit:125 - EPOCH 46000, loss=0.0000, lr=0.0004
2025-10-01 13:29:54.183 | INFO | main:fit:125 - EPOCH 47000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.217 | INFO | main:fit:125 - EPOCH 48000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.250 | INFO | main:fit:125 - EPOCH 49000, loss=0.0000, lr=0.0001
2025-10-01 13:29:54.285 | INFO | main:fit:125 - EPOCH 50000, loss=0.0000, lr=0.0001
2025-10-01 13:29:54.324 | INFO | main:fit:125 - EPOCH 51000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.355 | INFO | main:fit:125 - EPOCH 52000, loss=0.0000, lr=0.0004
2025-10-01 13:29:54.395 | INFO | main:fit:125 - EPOCH 53000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.424 | INFO | main:fit:125 - EPOCH 54000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.467 | INFO | main:fit:125 - EPOCH 55000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.501 | INFO | main:fit:125 - EPOCH 56000, loss=0.0000, lr=0.0002
2025-10-01 13:29:54.535 | INFO | main:fit:125 - EPOCH 57000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.575 | INFO | main:fit:125 - EPOCH 58000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.611 | INFO | main:fit:125 - EPOCH 59000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.647 | INFO | main:fit:125 - EPOCH 60000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.682 | INFO | main:fit:125 - EPOCH 61000, loss=0.0000, lr=0.0001
2025-10-01 13:29:54.716 | INFO | main:fit:125 - EPOCH 62000, loss=0.0000, lr=0.0005
2025-10-01 13:29:54.755 | INFO | main:fit:125 - EPOCH 63000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.790 | INFO | main:fit:125 - EPOCH 64000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.826 | INFO | main:fit:125 - EPOCH 65000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.855 | INFO | main:fit:125 - EPOCH 66000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.898 | INFO | main:fit:125 - EPOCH 67000, loss=0.0000, lr=0.0000
2025-10-01 13:29:54.933 | INFO | main:fit:125 - EPOCH 68000, loss=0.0000, lr=0.0008
2025-10-01 13:29:54.966 | INFO | main:fit:125 - EPOCH 69000, loss=0.0000, lr=0.0000
2025-10-01 13:29:55.005 | INFO | test_main:test_regression_gd:44 - model.weights=array([3.]) , model.intercept=np.float64(3.9999999999549973)
.
===== 2 passed in 3.21s =====

```

B.

```
● PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\hw1> uv run flake8 .\main.py
○ PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\hw1>

● PS C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\hw1> uv run pytest .\test_main.py -s
===== test session starts =====
platform win32 -- Python 3.12.2, pytest-8.4.2, pluggy-1.6.0
rootdir: C:\Users\percy\OneDrive\桌面\大學資料\3上\ML\hw1
configfile: pyproject.toml
collected 2 items

test_main.py (100, 1)
2025-10-01 13:22:58.605 | INFO | test_main:test_regression_cf:30 - model.weights=array([[3.]], model.intercept=array([4.])
(100, 1)
2025-10-01 13:22:58.606 | INFO | main:fit:89 - EPOCH 0, loss=30622.0606, lr=0.0001
2025-10-01 13:22:58.616 | INFO | main:fit:89 - EPOCH 1000, loss=13.0996, lr=0.0001
2025-10-01 13:22:58.624 | INFO | main:fit:89 - EPOCH 2000, loss=10.7249, lr=0.0001
2025-10-01 13:22:58.633 | INFO | main:fit:89 - EPOCH 3000, loss=8.7807, lr=0.0001
2025-10-01 13:22:58.640 | INFO | main:fit:89 - EPOCH 4000, loss=7.1890, lr=0.0001
2025-10-01 13:22:58.649 | INFO | main:fit:89 - EPOCH 5000, loss=5.8858, lr=0.0001
2025-10-01 13:22:58.654 | INFO | main:fit:89 - EPOCH 6000, loss=4.8188, lr=0.0001
2025-10-01 13:22:58.666 | INFO | main:fit:89 - EPOCH 7000, loss=3.9453, lr=0.0001
2025-10-01 13:22:58.674 | INFO | main:fit:89 - EPOCH 8000, loss=3.2301, lr=0.0001
2025-10-01 13:22:58.685 | INFO | main:fit:89 - EPOCH 9000, loss=2.6445, lr=0.0001
2025-10-01 13:22:58.695 | INFO | main:fit:89 - EPOCH 10000, loss=2.1651, lr=0.0001
2025-10-01 13:22:58.704 | INFO | main:fit:89 - EPOCH 11000, loss=1.7727, lr=0.0001
2025-10-01 13:22:58.713 | INFO | main:fit:89 - EPOCH 12000, loss=1.4513, lr=0.0001
2025-10-01 13:22:58.722 | INFO | main:fit:89 - EPOCH 13000, loss=1.1882, lr=0.0001
2025-10-01 13:22:58.731 | INFO | main:fit:89 - EPOCH 14000, loss=0.9728, lr=0.0001
2025-10-01 13:22:58.738 | INFO | main:fit:89 - EPOCH 15000, loss=0.7965, lr=0.0001
2025-10-01 13:22:58.749 | INFO | main:fit:89 - EPOCH 16000, loss=0.6521, lr=0.0001
2025-10-01 13:22:58.756 | INFO | main:fit:89 - EPOCH 17000, loss=0.5339, lr=0.0001
2025-10-01 13:22:58.765 | INFO | main:fit:89 - EPOCH 18000, loss=0.4371, lr=0.0001
2025-10-01 13:22:58.774 | INFO | main:fit:89 - EPOCH 19000, loss=0.3579, lr=0.0001
2025-10-01 13:22:58.780 | INFO | main:fit:89 - EPOCH 20000, loss=0.2930, lr=0.0001
2025-10-01 13:22:58.791 | INFO | main:fit:89 - EPOCH 21000, loss=0.2399, lr=0.0001
2025-10-01 13:22:58.800 | INFO | main:fit:89 - EPOCH 22000, loss=0.1964, lr=0.0001
2025-10-01 13:22:58.804 | INFO | main:fit:89 - EPOCH 23000, loss=0.1608, lr=0.0001
2025-10-01 13:22:58.815 | INFO | main:fit:89 - EPOCH 24000, loss=0.1316, lr=0.0001
2025-10-01 13:22:58.824 | INFO | main:fit:89 - EPOCH 25000, loss=0.1078, lr=0.0001
2025-10-01 13:22:58.830 | INFO | main:fit:89 - EPOCH 26000, loss=0.0882, lr=0.0001
2025-10-01 13:22:58.842 | INFO | main:fit:89 - EPOCH 27000, loss=0.0722, lr=0.0001
2025-10-01 13:22:58.853 | INFO | main:fit:89 - EPOCH 28000, loss=0.0591, lr=0.0001
2025-10-01 13:22:58.863 | INFO | main:fit:89 - EPOCH 29000, loss=0.0484, lr=0.0001
2025-10-01 13:22:58.873 | INFO | main:fit:89 - EPOCH 30000, loss=0.0396, lr=0.0001
2025-10-01 13:22:58.882 | INFO | main:fit:89 - EPOCH 31000, loss=0.0325, lr=0.0001
2025-10-01 13:22:58.884 | INFO | main:fit:89 - EPOCH 32000, loss=0.0266, lr=0.0001
2025-10-01 13:22:58.897 | INFO | main:fit:89 - EPOCH 33000, loss=0.0218, lr=0.0001
2025-10-01 13:22:58.904 | INFO | main:fit:89 - EPOCH 34000, loss=0.0178, lr=0.0001
2025-10-01 13:22:58.914 | INFO | main:fit:89 - EPOCH 35000, loss=0.0146, lr=0.0001
2025-10-01 13:22:58.924 | INFO | main:fit:89 - EPOCH 36000, loss=0.0119, lr=0.0001
2025-10-01 13:22:58.935 | INFO | main:fit:89 - EPOCH 37000, loss=0.0098, lr=0.0001
2025-10-01 13:22:58.943 | INFO | main:fit:89 - EPOCH 38000, loss=0.0080, lr=0.0001
2025-10-01 13:22:58.947 | INFO | main:fit:89 - EPOCH 39000, loss=0.0066, lr=0.0001
2025-10-01 13:22:58.960 | INFO | main:fit:89 - EPOCH 40000, loss=0.0054, lr=0.0001
2025-10-01 13:22:58.964 | INFO | main:fit:89 - EPOCH 41000, loss=0.0044, lr=0.0001
2025-10-01 13:22:58.977 | INFO | main:fit:89 - EPOCH 42000, loss=0.0036, lr=0.0001
2025-10-01 13:22:58.984 | INFO | main:fit:89 - EPOCH 43000, loss=0.0029, lr=0.0001
2025-10-01 13:22:58.992 | INFO | main:fit:89 - EPOCH 44000, loss=0.0024, lr=0.0001
2025-10-01 13:22:59.001 | INFO | main:fit:89 - EPOCH 45000, loss=0.0020, lr=0.0001
2025-10-01 13:22:59.009 | INFO | main:fit:89 - EPOCH 46000, loss=0.0016, lr=0.0001
2025-10-01 13:22:59.015 | INFO | main:fit:89 - EPOCH 47000, loss=0.0013, lr=0.0001
2025-10-01 13:22:59.027 | INFO | main:fit:89 - EPOCH 48000, loss=0.0011, lr=0.0001
2025-10-01 13:22:59.035 | INFO | main:fit:89 - EPOCH 49000, loss=0.0009, lr=0.0001
2025-10-01 13:22:59.042 | INFO | main:fit:89 - EPOCH 50000, loss=0.0007, lr=0.0001
2025-10-01 13:22:59.047 | INFO | main:fit:89 - EPOCH 51000, loss=0.0006, lr=0.0001
2025-10-01 13:22:59.060 | INFO | main:fit:89 - EPOCH 52000, loss=0.0005, lr=0.0001
2025-10-01 13:22:59.069 | INFO | main:fit:89 - EPOCH 53000, loss=0.0004, lr=0.0001
2025-10-01 13:22:59.078 | INFO | main:fit:89 - EPOCH 54000, loss=0.0003, lr=0.0001
2025-10-01 13:22:59.086 | INFO | main:fit:89 - EPOCH 55000, loss=0.0003, lr=0.0001
2025-10-01 13:22:59.096 | INFO | main:fit:89 - EPOCH 56000, loss=0.0002, lr=0.0001
2025-10-01 13:22:59.105 | INFO | main:fit:89 - EPOCH 57000, loss=0.0002, lr=0.0001
2025-10-01 13:22:59.114 | INFO | main:fit:89 - EPOCH 58000, loss=0.0001, lr=0.0001
2025-10-01 13:22:59.123 | INFO | main:fit:89 - EPOCH 59000, loss=0.0001, lr=0.0001
2025-10-01 13:22:59.130 | INFO | main:fit:89 - EPOCH 60000, loss=0.0001, lr=0.0001
2025-10-01 13:22:59.138 | INFO | main:fit:89 - EPOCH 61000, loss=0.0001, lr=0.0001
2025-10-01 13:22:59.147 | INFO | main:fit:89 - EPOCH 62000, loss=0.0001, lr=0.0001
2025-10-01 13:22:59.155 | INFO | main:fit:89 - EPOCH 63000, loss=0.0001, lr=0.0001
2025-10-01 13:22:59.164 | INFO | main:fit:89 - EPOCH 64000, loss=0.0000, lr=0.0001
2025-10-01 13:22:59.171 | INFO | main:fit:89 - EPOCH 65000, loss=0.0000, lr=0.0001
2025-10-01 13:22:59.181 | INFO | main:fit:89 - EPOCH 66000, loss=0.0000, lr=0.0001
2025-10-01 13:22:59.189 | INFO | main:fit:89 - EPOCH 67000, loss=0.0000, lr=0.0001
2025-10-01 13:22:59.203 | INFO | main:fit:89 - EPOCH 68000, loss=0.0000, lr=0.0001
2025-10-01 13:22:59.213 | INFO | main:fit:89 - EPOCH 69000, loss=0.0000, lr=0.0001
2025-10-01 13:22:59.215 | INFO | test_main:test_regression_gd:44 - model.weights=array([3.]), model.intercept=np.float64(3.9963537486342404)
+
===== 2 passed in 1.26s =====
```

Part. 2, Questions (40%):

- (10%) Linear models $y = w^T x + b$ have limited fitting power.
 - In one sentence, explain why a single linear model is limited.

- b. Give one concrete task that a single linear model cannot solve, and state why no single hyperplane/affine function solves it.

[a. Single linear model can only use single hyperplane to make data regression or classification, it can not solve the problem that are not linear separable like a circle around a dot at center.]

[b. A circle around a dot at center. In classification, let the center dot be a class, and the circle are another class, then no hyperplane can separate them; In regression, there is no hyperplane to reduce the losses because in all direction, they all the same.]

- 2. (15%) Why do we add a regularization term in linear regression? What are the differences between L2 regularization (Ridge) and L1 regularization (Lasso)? Please explain in detail.

[Add a regularization term (penalty term) to the loss function can discourage the weights from reaching large values, which avoid overfitting.]

[L2 regularization (Ridge) sums up the square of the coefficient, L1 sums up the absolute value of coefficient. L1 encourages the model to use fewer coefficients, reducing the number of features used in the model, make model would not be “too complicate” ; L2 encourages the model to use smaller coefficients, make model would not focus on specific feature. If there is a coefficient too large, L1 would probably make it become 0 but L2 would pretend to make it smaller. If want sparse (specific feature choosing) -> L1, want smoother -> L2.]

- 3. (15%)

- What is overfitting? Under what conditions can a model overfit? (List two) How can overfitting be alleviated? (List two)

[Model is extremely accurate on the training data but perform poorly on the test data.]

[Model is too complicated and the data is not clean enough, too many features and parameters make model can learn noises, not the rule of the data;

the training data is too small, there is not enough data or features for model to learn the rule]

[Add regularization term, to alleviate the problem of the large absolute value of weight, like L2 regularization (Ridge);

Data augmentation, to alleviate the problem of training data not diverse enough]